Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/08/2007
PI Name:	Kennedy, Ann R. D.Sc.		
Project Title:	Countermeasures for Space Radiation Induced Myeloid Leukemia		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Radiation Effects Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	19104-6072	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	06/01/2004	**End Date:**	09/30/2007
No. of Post Docs:	2	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	2
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Gewirtz, Alan (University of Pennsylvania) Biaglow, John (University of Pennsylvania) Ware, Jeffrey (University of Pennsylvania) Stewart, Jelena (University of Pennsylvania) Newberne, Paul (Boston Medical Center)		
Grant/Contract No.:	NCC 9-58-IIH00402		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This proposed research investigates mechanisms to resist malignancy in space with an emphasis on evaluation of certain nutritional supplements as countermeasures to protect against space radiation-induced leukemogenesis; these nutritional supplements include L-selenomethionine (SeM), vitamin C, vitamin E succinate, alpha-lipoic acid, Co-enzyme Q10 and N-acetylcysteine. Data obtained from our previous in vitro and animal studies supported by the National Space Biomedical Research Institute (NSBRI) have demonstrated that the nutritional supplements to be evaluated in this proposal are effective in preventing radiation induced oxidative stress and protecting cultured human cells against adverse biological effects induced by the types of radiation of most concern during space travel, i.e., highly energetic heavy charged particles (known as HZE particles) and protons. This proposed investigation will determine the efficacy of a combination of these nutritional supplements as a countermeasure against space radiation induced acute myeloid leukemia (AML) using the CBA mouse model system. Two 2-year animal experiments are planned to determine the effect of supplement combination on the development of AML in CBA mice exposed to radiation with HZE particles (1-GeV or 5-GeV/nucleon iron ions) or protons; blood cell profiles and gene expression (of myeloperoxidase [MPO] in the spleen and CD33 in the bone marrow) patterns will be monitored throughout the studies, and gene expression patterns of the oxidation resistance gene (OXR1) will be determined at the end of the studies. In addition, several short-term animal and cell culture experiments will be performed to determine the effects of radiation treatment and nutritional supplementation on three selected surrogate endpoint biomarkers (SEBs), which are the host bio-reduction capacity measured as plasma total antioxidant status, plasma protein carbonyl content and OXR1 gene expression in selected populations of white blood cells. The results of this proposed study are expected to provide critical information about the feasibility and mechanism(s) of nutritional supplements in increasing the resistance to space radiation induced malignancy. In the past year, we have continued our long-term studies on the induction of AML by protons and HZE particles in CBA mice and performed several cell culture experiments. In the next year of the grant, animals will continue to be monitored as part of the long-term animal studies described in the grant and more short-term animal and cell culture experiments will be performed.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	People living on earth are exposed to higher than normal doses of radiation from a variety of sources and for a variety of reasons. It is expected that the countermeasures being developed as part of this NSBRI research project will be equally useful for the protection of people on earth from the expected adverse effects of exposure to ionizing radiation.
Task Progress:	This research program involves studies of nutritional supplements as countermeasures to protect against space radiation-induced leukemogenesis. As part of this research, we are determining the efficacy of a combination of nutritional supplements as a countermeasure against space radiation (from protons and HZE particles) induced acute myeloid leukemia (AML) using the CBA mouse model system. In addition, several cell culture experiments are being performed to determine the effects of radiation treatment and nutritional supplementation on selected surrogate endpoint biomarkers (SEBs) involved in radiation induced malignancy. In this past year, much effort has been focused on studies performed to determine whether changes in gene expression can be used as SEBs in this program. Microarray technology has been used to assess the effects of an antioxidant (Aox) combination (containing L-selenomethionine, a-lipoic acid, vitamin E succinate, N-acetyl cysteine and ascorbic acid) on HZE particle radiation-induced changes in gene expression. CRL-9855 cells (human mononuclear cells) were exposed to two low radiation doses: 10 cGy or 20 cGy from iron ions, in the presence or in the absence of the Aox combination, and were harvested for RNA analysis 6 and 16 hours post irradiation. U133Av2 Gene Chips from Affymetrix were used to assess changes in expression of over 22,000 targets. Protocols were conducted as described in the Affymetrix GeneChip Expression Analysis Technical Manual. Significant differences in gene expression levels were determined using a log scale robust multi-array analysis (RMA). Triplicate expression values that resulted from using RMA were further analyzed using a 3-way ANOVA; variables were radiation dose, agent treatment and time post-irradiation. Differentially expressed genes were further analyzed using EASE overrepresentation. When lists of significantly regulated genes were compared, it is apparent that the Aox combination had the greatest effect on early changes in radiation induced gene expression. In cells exposed to a 10 cGy radiation dose, regulation of 173 out of 187 genes was modified in the presence of the Aox combination. Similarly, in cells exposed to a 20 cGy radiation dose, regulation of 574 out of 597 genes was modified in the presence of the Aox combination. This is an impressive change in gene regulation due to supplementation with the Aox combination which could account for a protective effect of the Aox combination against radiation induced biological effects. The EASE overrepresentation analysis indicates that a significant number of genes involved in the apoptotic pathway are regulated in the early response to radiation exposure from either 10 or 20 cGy doses. In cells irradiated in the presence of the Aox combination, radiation exposure does not induce regulation of transcription of these genes. This is a very exciting observation which we are investigating in further detail.
Bibliography Type:	Description: (Last Updated: 08/24/2020)
Articles in Peer-reviewed Journals	Guan J, Stewart J, Ware JH, Zhou Z, Donahue JJ, Kennedy AR. "Effects of dietary supplements on the space radiation-induced reduction in total antioxidant status in CBA mice." Radiat Res. 2006 Apr;165(4):373-8. PMID: 16579649 , Apr-2006
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Articles in Peer-reviewed Journals	Wan XS, Zhou Z, Ware JH, Kennedy AR. "Standardization of a fluorometric assay for measuring oxidative stress in irradiated cells." Radiat Res. 2005 Feb;163(2):232-40. PMID: 15658900 , Feb-2005